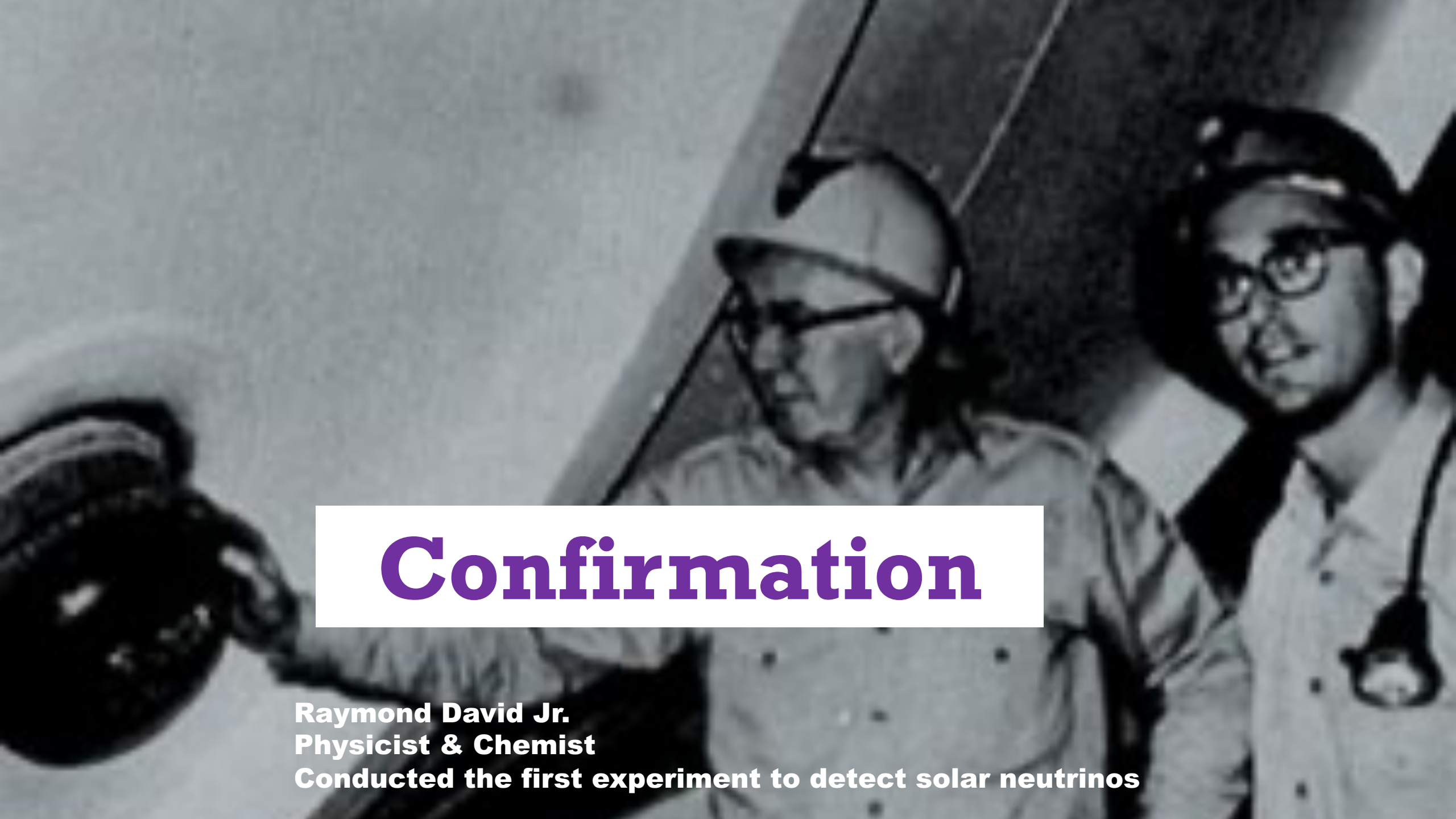


A black and white photograph of two men in a laboratory. The man on the left, Raymond David Jr., is wearing a hard hat and safety glasses, and is holding a large, dark, circular object. The man on the right is wearing a hard hat and glasses, and is looking towards the camera. The background shows industrial equipment and cables.

Confirmation

Raymond David Jr.
Physicist & Chemist

Conducted the first experiment to detect solar neutrinos

ANNOUNCEMENT

Assignment: Role Play of Scientific Figure

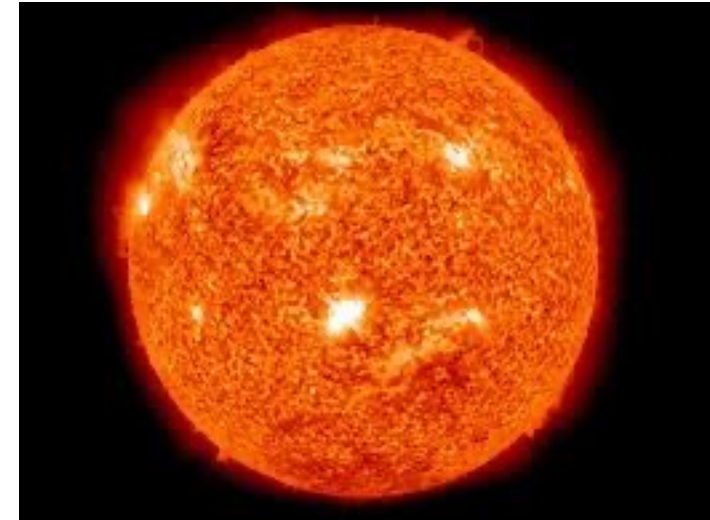
Deliverable 1: Name of the Scientific Figure – due to Jan 28th, 11:59 pm

Deliverable 2: Script – due to Feb 6th, 11:59 pm

Deliverable 3: Full Submission – due to Feb 15th, 11:59 pm

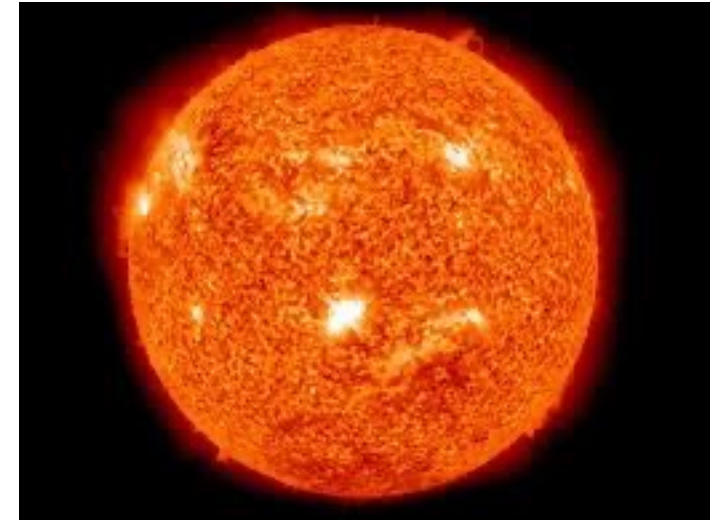
CONFIRMATION

- We find many *competing* claims about unseen events and objects that explain the phenomena we see:
 - Example: Where does the Sun's energy come from?
 - From the Greek deity Helios.
 - The Sun is a burning ball of coal or fuel.
 - The Sun is slowly shrinking due to gravity, converting gravitational energy into heat and light.
 - Radioactivity is the Sun's energy source.
 - The Sun's energy source is an ongoing process of nuclear fusion, which showers us with heat, light, and neutrinos.



CONFIRMATION

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But at most one of those theories is true (*likely to be true*).

CONFIRMATION

- The responsibility of science is to weed out the 'bad' theories from the 'good'.
- **How?**
 - By testing the theory against the observable data.

CONFIRMATION

- The responsibility of science is to weed out the ‘bad’ theories from the good.
- **How?**
 - By testing the theory against the observable data.



But what do we mean by
“testing” a theory and “by
comparison with the
observable evidence”?

CONFIRMATION

Example:

- Theoretical claim: “Helium atoms contain two electrons.”
- Checking is not a matter of looking around and counting (we cannot directly observe these things!).
- Rather, it is a matter of *checking on the consequences* of the theoretical claim by determining what *effect* the electrons must have on the *observable world*.

Last time!

INTRODUCTION

When we “reduce” electron to its experiential consequences, we get something like bubbles in a cloud chamber. We don’t see electrons directly. We see the bubble trail it leaves.



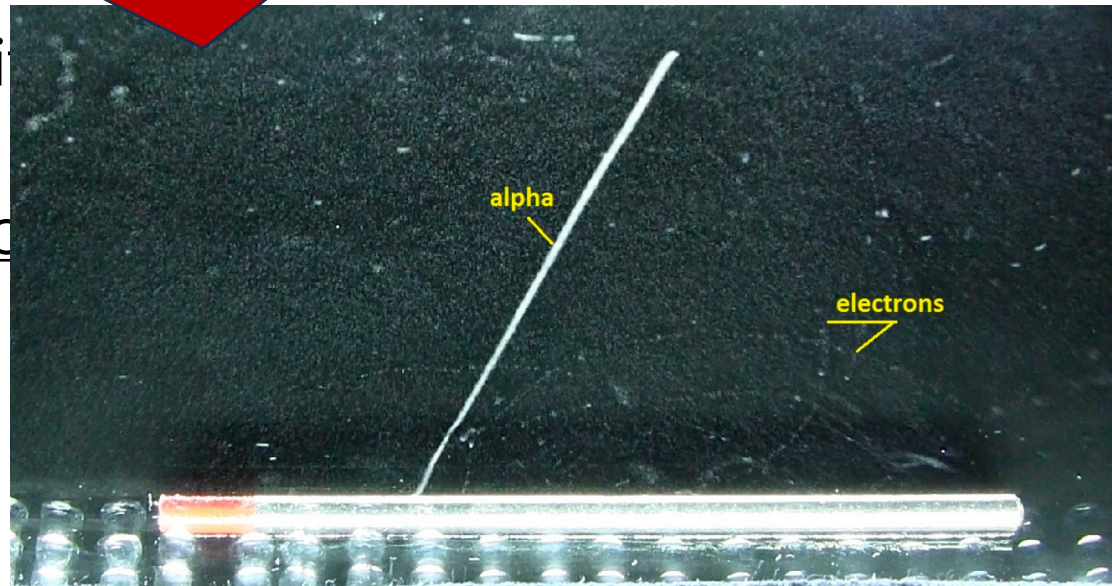
Cloud Chamber

heat, etc. The phenomena is observable.

- But how can we verify

directly

gain two



Last time!

INTRODUCTION

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Cloud Chamber

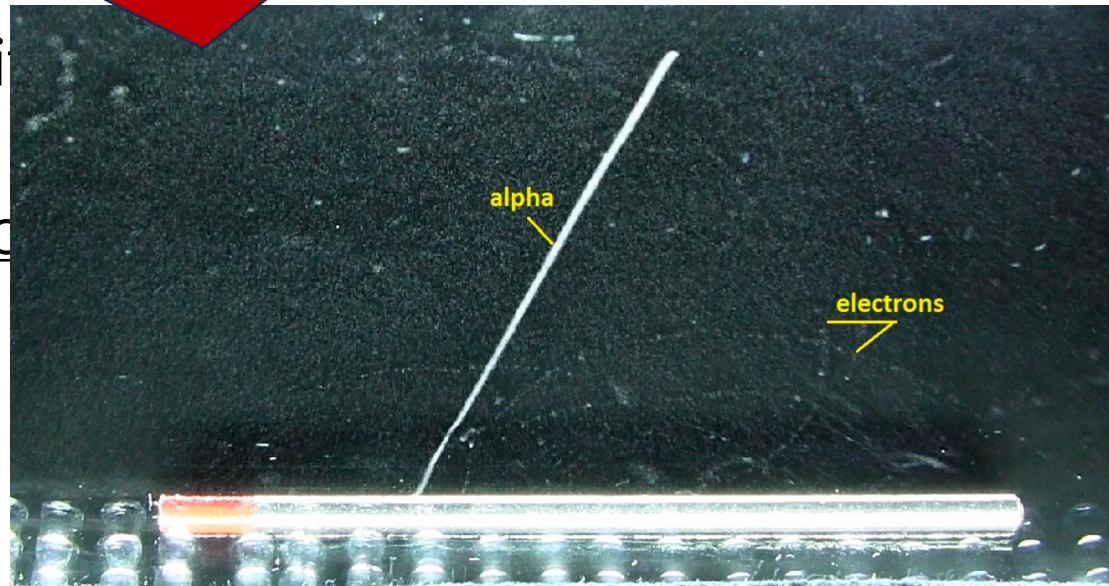
heat, etc. The phenomena is observable.

verifi

tly c

ain two

There is a kind of **indirectness** in any case of using observational evidence to test a theory.



CONFIRMATION

- However,:
 - A theory *must* be accountable to *observational justification*.
 - This means that not only must a theory be **testable** (internal virtue), but it must be also be **tested**.

CONFIRMATION

- Example: Where does the Sun's energy come from?
 - The Sun's energy source is an ongoing process of nuclear fusion, which showers us with heat, light, and neutrinos.
 - And some of the neutrinos ought to find their way to the earth where they can be detected.
 - The theory is testable since the neutrinos are detectable.



Raymond David Jr. detected solar neutrinos by using a 100,000-gallon tank of chlorine-rich cleaning fluid placed 1.5 km underground in a gold mine. Neutrinos occasionally interact with the chlorine, changing it into a radioactive argon atom. The radioactive emissions from the argon can then be counted as the footprints of solar neutrinos.

CONFIRMATION



Reflect & Discuss

1. *“...in a sense there is no rigid dichotomy between the activities of confirmation and explanation ...they follow the same logical form, which compares the consequences of theories to observations of the world and thus connects theories with the world by observing the effects that the theorized entities have on objects we can see.”*

The text suggests that explanation and confirmation are two sides of the same coin. Are they different in any way?

CONFIRMATION

- Confirmation and explanation follow the *same logical form*:
 - Both *compare the consequences of a theory to observations in the world* and thus connect theories with the world by observing the effects that the theorized entities have on objects we can see.
 - Example: *Solar Theory*
 - If the neutrinos are detected we will say that they contribute to the *confirmation* of the solar theory.
 - And that the solar theory *explains* the shower of neutrinos.

CONFIRMATION

- It is typically, though,
 - to apply the concept of ***explanation*** to ***past observations*** (things we knew happened but wondered why),
 - and reserve the ***concept of testing*** to cover ***future observations***, predictions whose outcome is yet unknown.

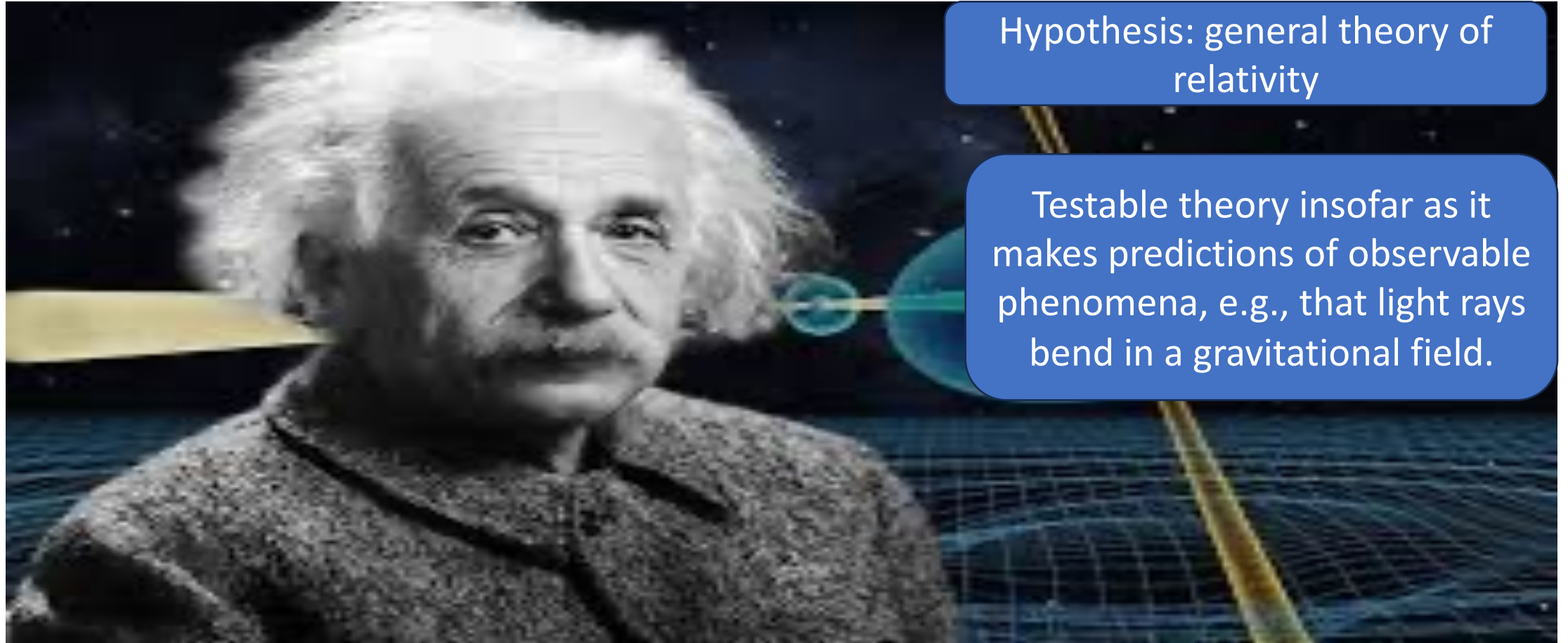
CONFIRMATION

- Explanations and confirmation are also distinguished by the pragmatics of the situation:
 - If it is the phenomenon itself that is puzzling and the question is “***Why does this happen?***” then it is ***explanation*** scientists are after.
 - If the interest is centered on the ***status of the theory itself*** and the phenomena are just hoops it must jump, then it is ***confirmation***.

CONFIRMATION

- Our focus in today's class is not so much on *why* the phenomena happened but rather *that* they happened as predicted by the theory.

CONFIRMATION



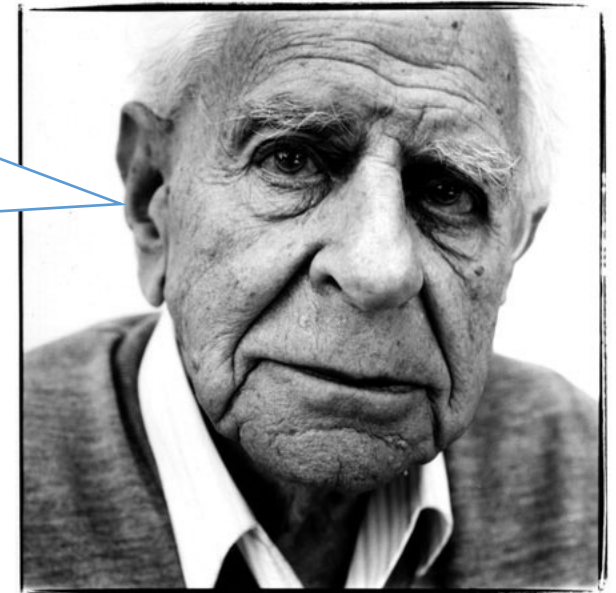
Hypothesis: general theory of relativity

Testable theory insofar as it makes predictions of observable phenomena, e.g., that light rays bend in a gravitational field.

A BASIC MODEL OF CONFIRMATION

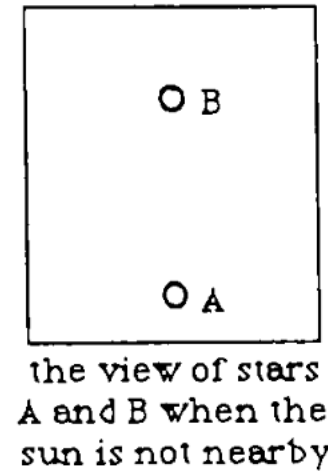
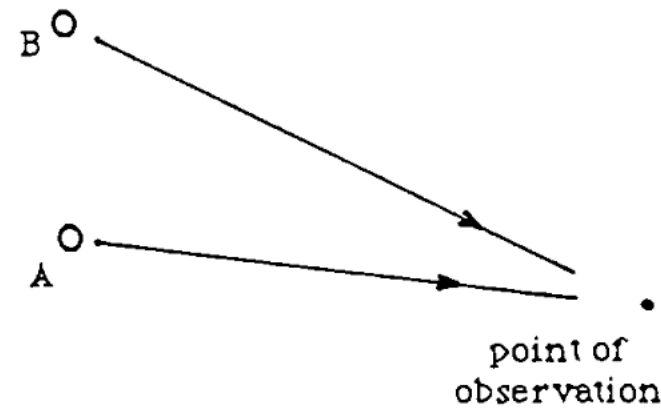
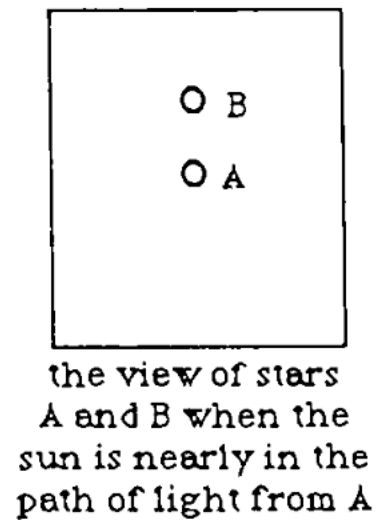
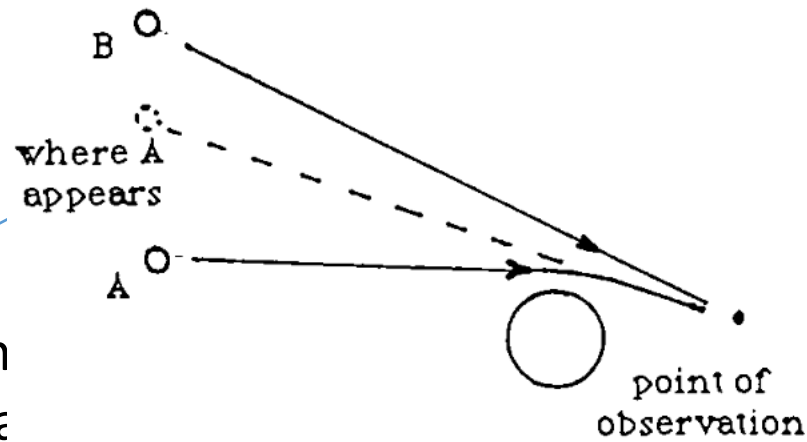
“Einstein’s general theory of relativity entails that light rays must bend in a gravitational field. Organized by Sir Arthur Eddington, two Royal Astronomical Society expeditions were dispatched to observe the solar eclipse of 1919, and verified that starlight was indeed deflected by the sun by the amount that Einstein had predicted. The Times of London reported the success as the most remarkable scientific event since the discovery of the planet Neptune.”

- **Einstein’s general theory of relativity
put to the test**



Sir Karl Popper

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of Confirmation:

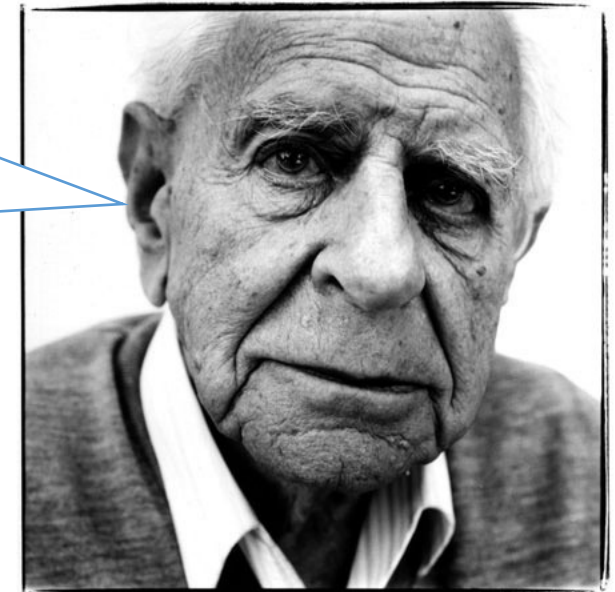


Figure 4.1. Two stars, A and B, viewed in two circumstances to see if the gravity of the sun affects the trajectory of light.

A BASIC MODEL OF CONFIRMATION

- The prediction was confirmed, but was the theory? What's the difference?
- What does one true prediction prove?

A BASIC MODEL OF CONFIRMATION

Hypothetico-deductive confirmation (H-D):

- This is a very general, formal model of confirmation in which a *hypothesis* is tested by its *deductive consequences*.
- It is by no means a deductive proof that the hypothesis is true.
- Rather, it is a way of providing *inductive* evidence that the hypothesis is *likely to be true*.

For **example**, the hypothesis that all dogs are vicious has the deductive consequence that the collie next door is vicious. I test this by testing the disposition of the collie next door. If she is vicious, then this is **some small** indication that the hypothesis is true.



A BASIC MODEL OF CONFIRMATION

- If one successful H-D test helps a little, further H-D tests will help more.

Example:

- The theory of relativity does not rest its credibility on the single success of predicting the bending of starlight.
- It makes other predictions, for example, that gravity affects the passage of time and the frequency of light. This is the gravitational red shift, and it too has been observed (e.g., clocks on Earth tick more slowly than those in satellites).

A BASIC MODEL OF CONFIRMATION

- The prediction was confirmed, but was the theory? What's the difference?
- What does one true prediction prove?
- Suppose the prediction turned out to be wrong?

A BASIC MODEL OF CONFIRMATION



Reflect & Discuss

2. *“By virtue of the logical pattern of drawing predictions from theories, falsification has the advantage of a stricter standard, a deductive demonstration of failure.”*

Explain

A BASIC MODEL OF CONFIRMATION

- In the case, where the prediction fails, we speak of ***hypothetico-deductive falsification***:
 - The idea is that if the theorized object or event is *always* supposed to cause the observable effect mentioned, but in fact the effect is missing, then the theorized object must not be there.

A BASIC MODEL OF CONFIRMATION

- Note, there is an ***asymmetry*** in this model of testing between confirmation and falsification.
 - ***Confirmation*** comes *in degrees* and is vulnerable to confusion as to how much is enough.
 - ***Falsification***, on the other hand, shows no such sloppiness, no degrees, no need for discretion.
 - A single failed prediction will suffice to falsify a theory and force its rejection.

A BASIC MODEL OF CONFIRMATION

- The prediction was confirmed, but was the theory? What's the difference?
- What does one true prediction prove?
- Suppose the prediction turned out to be wrong?
- Would that have been sufficient evidence to disqualify the theory once and finally?

A BASIC MODEL OF CONFIRMATION

- Example: Where does the Sun's energy come from?
 - The Sun's energy source is an ongoing process of nuclear fusion, which showers us with heat, light, and neutrinos.
 - And some of the neutrinos ought to find their way to the earth where they can be detected.
 - The theory is testable since the neutrinos are detectable.
 - **Problem**: Discrepancy between the number of predicted neutrinos and the number Raymond David Jr. measured (known as ***Solar-Neutrino Problem***)



Raymond David Jr. detected solar neutrinos by using a 100,000-gallon tank of chlorine-rich cleaning fluid placed 1.5 km underground in a gold mine. Neutrinos occasionally interact with the chlorine, changing it into a radioactive argon atom. The radioactive emissions from the argon can then be counted as the footprints of solar neutrinos.

• The solar-neutrino problem is not a big enough problem to force the rejection of the theory.

In other words, the failure to observe the predicted effect has not been taken as conclusive proof that the theory is false.

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• The solar-neutrino problem is not a big enough problem to force the rejection of the theory.

In other words, the predicted value is taken as conclusive evidence that the theory is false.

It also doesn't take the experimental setup (e.g., the fluid may be contaminated) and the reasoning process that issued the prediction into account.

Discrepancy between the number of detected neutrinos and the number Raymond David Jr. measured (known as **Solar-Neutrino Problem**)



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A BASIC MODEL OF CONFIRMATION

- So it is more complicated as the simple H-D model indicates, ...
 - (e.g., a two-component deduction from the hypothesis about solar fusion to the prediction of clicks on the detector counting radioactive argon.)
- ... and a realistic model of scientific testing must take the details seriously.
 - (e.g., the conditions of testing.)

MORE DETAILED & REALISTIC MODEL OF CONFIRMATION



Reflect & Discuss

3. The revised H-D model has three components.

List them and explain the connection of each of them in excerpt 2 (smallpox vaccine).

MORE DETAILED & REALISTIC MODEL OF CONFIRMATION

- The *three components* of the revised ***hypothetico-deductive model*** are
 1. the hypothesis (that is, the theory being tested),
 2. the conditions of testing, and
 3. the predicted effect.
- If the hypothesis is true, then if conditions are right, the effect will be observed.
- If, in fact, the effect is not observed, then we are forced to conclude that ***either*** the hypothesis is false ***or*** the conditions are not right.

MORE DETAILED & REALISTIC MODEL OF CONFIRMATION



Reflect & Discuss

Read the excerpt 1 (the predicted orbital positions of the planet Uranus), take notes of any important points and definitions

4. Kosso describes the above episode in the following words “Here is a final but classic example of retaining a hypothesis in the face of failed prediction. Not only were auxiliaries doubted before the hypothesis, but a new, ad hoc auxiliary theory was devised to account for the unfavourable experimental outcome. The whole affair had a very profitable result .”

What is an auxiliary theory? Identify the auxiliaries in excerpt 1.

MORE DETAILED & REALISTIC MODEL OF CONFIRMATION

- In addition, it becomes clear that more than one theoretical claim is needed to deduce the predictions.
- The hypothesis alone does not entail the prediction.
- It is rather the hypothesis in conjunction with ***auxiliary theories***,
 - i.e., *the background knowledge required to draw testable consequences from a hypothesis*
- from which predictions are deduced.

MORE DETAILED & REALISTIC MODEL OF CONFIRMATION

- No hypothesis alone entails testable predictions, so auxiliary theories always participate in the testing of a hypothesis.
 - For example, the hypothesis that the solar interior is a furnace of nuclear fusion has the testable consequence that solar neutrinos are showering the earth, but the consequence can be drawn out only with the theoretical knowledge that fusion produces neutrinos. The claim about the production of neutrinos is an auxiliary theory in this case of testing.

MORE DETAILED & REALISTIC MODEL OF CONFIRMATION



Reflect & Discuss

5. In this class, you have gone through two case studies, which presented the process of scientific discovery in different disciplines: medicine and astronomy. Drawing from these examples, what characteristics and ways of thinking do you think enabled these scientists to make these significant advances?

MORE DETAILED & REALISTIC MODEL OF CONFIRMATION



Reflect & Discuss

6. “The observable evidence, as it is sometimes said, underdetermines the theoretical account of the world.”

What might this mean?